



CLOSING THE THREE-GAP FRAMEWORK: A SYSTEMATIC REVIEW OF EVALUATION, GOVERNANCE, AND IMPLEMENTATION CHALLENGES IN CLINICAL AI DEPLOYMENT *

Master's Thesis

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ABSTRACT

Despite rapid growth in research on artificial intelligence in healthcare, the literature remains fragmented across disciplinary silos that each address isolated facets of how AI is evaluated and governed, with little synthesis of how these dimensions interact in real-world deployment. This review identifies a three-gap framework as its central conceptual contribution, highlighting three interlocking structural deficits that constrain the safe and equitable translation of clinical AI from pilot projects to routine care: the **ecological validity gap** between controlled validation studies and the realities of clinical practice; the **governance capacity gap** between the institutional preconditions emerging regulatory frameworks assume and the organizational readiness health systems possess; and the **pilot trap**—the systematic failure of clinically-validated AI systems to transition from short-term demonstration projects into sustained components of routine care, often mediated by post-market performance drift and absent accountability infrastructure for continuously-learning systems. A systematic search was conducted across PubMed, Semantic Scholar, arXiv, and DBLP for peer-reviewed publications and preprints published between 2020 and 2026, yielding a final corpus of 311 papers that were synthesized thematically. The synthesis reveals that these three gaps are not independent. Rather, they reinforce one another in a self-perpetuating cycle: ecologically weak validation enables deployment into institutions lacking sufficient governance capacity, which in turn are unable to sustain the post-deployment monitoring and accountability structures that the pilot trap demands. Current evaluation practices remain dominated by retrospective benchmarking on curated datasets; participation of patients, clinicians, and communities in AI governance remains largely episodic rather than institutionalized; and the most promising regulatory initiatives are those that treat evaluation, governance, and implementation as components of a continuous lifecycle rather than as separate phases. A growing body of literature on ecologically grounded evaluation protocols, governance maturity models, and post-deployment monitoring frameworks suggests that this cycle can be broken, but only through integrated approaches that connect technical evaluation, institutional governance, and implementation practice. The review argues that safe and equitable health AI deployment requires moving beyond pre-market authorization alone. It calls for (i) continuous post-market performance monitoring using standardized real-world metrics; (ii) institutional capacity-building as a prerequisite for sustainable AI integration; and (iii) institutionalized participatory mechanisms that ensure AI systems reflect the needs and values of the communities they serve. The review concludes by outlining the most pressing unanswered questions at the evaluation–governance–implementation interface and proposes a research agenda for closing the three identified gaps.

Keywords: health AI, AI evaluation, algorithmic governance, clinical AI deployment, adaptive regulation, participatory governance, implementation science, post-market surveillance, ecological validity

1 Introduction

1.1 Background and Motivation

Health artificial intelligence has transitioned from a research curiosity to a clinical reality. As of 2024, the United States Food and Drug Administration (FDA) had authorized nearly 1,000 AI-enabled medical devices spanning

diagnostic radiology, cardiac monitoring, ophthalmology, and oncology applications [1]. The United Kingdom's National Health Service (NHS) has similarly deployed AI tools across multiple trusts, including AI-assisted diabetic retinopathy screening programs that have screened over one million patients and algorithmic triage systems embedded within primary care pathways [nhs2024ai, price2024nhs]. In the United States, Epic Systems — one of the largest electronic health record vendors — has integrated large language models into its ambient documentation and clinical inbox summarization platforms, deploying these tools across hundreds of hospital systems and raising novel questions about institutional liability, data governance, and continuous performance monitoring [epic2024ilm, 2]. The scale of deployment is accelerating: AI-powered digital health applications attracted over USD 20 billion in venture capital funding in 2023 alone, and major hospital networks in North America, Europe, and East Asia have established dedicated AI governance committees within the past three years.

Despite this rapid uptake, a persistent pattern has emerged in the literature. Clinical AI applications frequently succeed in pilot studies but fail to become sustained components of routine clinical care. In practice, this pattern takes a recognisable shape: an AI system may demonstrate strong technical performance and promising clinical outcomes during a short-term trial, only to be quietly abandoned, partially scaled, or never integrated into routine workflows once the pilot concludes — a phenomenon that Tian et al. [3] term the **pilot trap**. This gap between demonstrated efficacy in controlled settings and real-world effectiveness is not primarily a technical problem. As the reviewed literature reveals, it is deeply rooted in governance deficits: unclear accountability structures, absent post-market monitoring mechanisms, insufficient organizational capacity, and regulatory frameworks designed for static medical devices that cannot accommodate continuously evolving AI systems [4, 5]. The consequence is a growing corpus of deployed AI systems whose long-term clinical impact remains unevaluated, whose failure modes are poorly documented, and whose governance rests on assumptions of institutional readiness that have not yet been met in most health systems.

1.2 Scientific Context and Knowledge Gap

A growing body of literature has examined individual facets of health AI deployment. Several reviews have addressed **AI evaluation methodologies** in isolation, focusing on metric design, benchmarking datasets, and technical validation protocols [see e.g. 5]. Others have examined **regulatory approaches** to AI, including the FDA's Software Pre-Certification (Pre-Cert) Program and the European Union's AI Act [1]. Still others have explored **ethical and participatory dimensions** of clinical AI, emphasizing the need for community engagement, transparency, and accountability [6]. Recently, Hussein et al. [7] proposed a maturity model for AI governance in healthcare, offering a structured assessment framework that maps organizational readiness across multiple governance dimensions. Similarly, palama2026auditing presented a four-layer auditing and monitoring framework spanning bias detection, explainability, performance surveillance, and regulatory compliance, linking continuous monitoring to institutional decision-making hierarchies.

However, no single review has yet synthesized these three streams — **evaluation methods, governance structures, and implementation outcomes** — through a unified analytical lens that takes seriously the specific challenges of real-world clinical deployment. Existing reviews tend to treat evaluation as a purely technical exercise and governance as a purely regulatory exercise, failing to account for the organizational, social, and temporal dynamics that determine whether an AI system ultimately benefits patients. Tian et al. [3] have described this disconnect explicitly, demonstrating through an 18-month longitudinal study in China that the failure of AI systems to progress from pilot to routine care is attributable not to algorithmic inadequacy but to institutional bottlenecks: fragmented accountability, underfunded post-deployment infrastructure, and governance mechanisms that are misaligned with the iterative nature of adaptive AI systems. This review addresses that gap by systematically interrogating the intersection of evaluation practice, governance architecture, and implementation evidence in health AI.

1.3 Research Question and Objectives

The central research question guiding this review is:

What methods and frameworks exist for evaluating health AI systems in real-world deployment contexts, and how do governance structures shape their implementation, sustainability, and societal impact?

This question is operationalized through three sub-questions:

- (I) **Evaluation methods:** Which evaluation frameworks, metrics, and validation methodologies are used to assess the performance, reliability, robustness, and safety of clinical AI systems in post-deployment settings? How do these methods address bias detection, algorithmic drift, and generalizability across diverse patient populations?

- (II) **Governance structures:** What regulatory, organizational, and participatory governance mechanisms have been proposed or implemented to ensure accountability, transparency, and equity in clinical AI deployment? How do these mechanisms interact across the regulatory lifecycle (pre-market authorization, post-market surveillance, adaptive updates)?
- (III) **Implementation outcomes and barriers:** What are the documented barriers to sustainable clinical AI implementation, and what organizational, technical, and social factors determine whether AI systems transition from pilot projects to routine care? What does the evidence say about real-world clinical impact?

1.4 Scope and Delimitation

This review focuses on peer-reviewed empirical and conceptual literature published between 2020 and 2026, covering AI systems deployed in or evaluated for direct or indirect clinical care — including diagnostic AI, predictive modeling, clinical decision support, digital phenotyping, and AI-assisted therapeutic applications. Studies limited to pre-clinical or animal model settings, or focusing exclusively on administrative optimization without direct clinical relevance, are excluded. The review further delimits its scope to human health AI applications, thereby excluding veterinary, environmental, or non-clinical AI deployments, as these latter domains involve distinct regulatory regimes, evaluation criteria, and societal implication profiles that fall outside the scope of the present analysis.

The review draws on a curated corpus extracted from PubMed, Semantic Scholar, arXiv, and Google Scholar using a keyword set organized across five thematic axes: *AI Evaluation and Validation*, *Participatory Governance*, *Adaptive Regulation*, *Evidence and Implementation*, and *Clinical Health AI*. A full description of the search strategy and selection criteria is provided in Section 2.

1.5 Report Structure

The remainder of this report is organized as follows. Section ?? situates this review within the existing literature on health AI evaluation and governance. Section 2 describes the systematic search strategy, selection criteria, and analytical approach. Section 3 presents the synthesized findings across the three analytical dimensions defined above. Section 4 discusses the implications of these findings for research, policy, and clinical practice, and articulates the key limitations of this review. Section ?? offers a concluding synthesis and a research agenda. Supplementary materials, including the full PRISMA flow diagram and the annotated bibliography, are provided in Appendix ??.

2 Methods

This section has two components. First, it describes the automated literature collection pipeline built to support this review — the technical and conceptual architecture that enabled systematic coverage of a rapidly evolving literature. Second, it describes the systematic review methodology itself: the search strategy, selection criteria, quality assessment, and analytical framework applied to the gathered corpus.

2.1 Pipeline Design and Implementation

The search was conducted via a semi-automated AI-assisted pipeline querying four databases (arXiv, PubMed, Semantic Scholar, DBLP) daily across five thematic axes spanning AI evaluation, participatory governance, adaptive regulation, evidence and implementation, and clinical health AI. Papers were scored using a multi-dimensional composite scoring algorithm and prioritized for human review. Full technical specifications — including the pipeline architecture, scoring dimensions, deployment scripts, and Telegram integration — are provided in Appendix ??.

2.2 Search Strategy and Information Sources

For the systematic review itself, papers were retrieved from the pipeline database covering the period January 2020 to April 2026. The search was replicated across four electronic databases to ensure comprehensive coverage: **PubMed** (MEDLINE), **Semantic Scholar**, **arXiv**, and **Google Scholar**. The search query was constructed around the five thematic axes described above, each comprising controlled vocabulary terms and free-text keywords:

- (I) **AI Evaluation and Validation** — “AI evaluation,” “AI validation,” “algorithmic auditing,” “model evaluation,” “bias detection,” “robustness testing,” “post-deployment monitoring,” “continuous evaluation,” “explainability,” “safety evaluation.”
- (II) **Participatory Governance** — “participatory governance,” “stakeholder engagement,” “citizen participation,” “algorithmic accountability,” “democratic AI,” “community-centered AI.”

- (III) **Adaptive Regulation** — “adaptive governance,” “agile regulation,” “responsive regulation,” “regulatory sandboxes,” “living labs,” “evidence-based regulation,” “continuous compliance.”
- (IV) **Evidence and Implementation** — “evidence-based policy,” “implementation science,” “knowledge translation,” “policy uptake,” “evidence gaps.”
- (V) **Clinical Health AI** — “clinical AI,” “medical AI,” “clinical decision support,” “diagnostic AI,” “predictive modeling in healthcare,” “digital health,” “AI safety in healthcare.”

Search strings were adapted for each database using Boolean operators (AND/OR) and database-specific syntax. No language restrictions were applied at the search stage; however, only English-language publications were retained after abstract screening. Preprints and peer-reviewed articles were both included given the rapid evolution of the field.

2.3 Selection Criteria

Studies were included if they met all of the following criteria:

- **Population:** Human health AI systems deployed in clinical, sub-clinical, or community health settings; or AI systems evaluated for direct or indirect clinical application.
- **Intervention:** Any AI-based intervention — including diagnostic AI, predictive modeling, clinical decision support systems, natural language processing for clinical text, wearable sensor analytics, and AI-assisted therapeutic applications.
- **Outcomes:** Studies reporting on evaluation metrics, governance structures, implementation barriers, regulatory frameworks, ethical considerations, or societal impact of health AI.
- **Study design:** Peer-reviewed empirical studies (RCTs, cohort studies, case studies, cross-sectional studies), systematic reviews, scoping reviews, conceptual papers, and policy analyses. Animal studies, purely computational studies without clinical application, and publications in predatory journals were excluded.
- **Date range:** Published between 2020 and 2026.

2.4 Quality Assessment

Given the heterogeneous study designs included in this review, a modular quality appraisal approach was adopted. For empirical clinical studies, the **JBIC Critical Appraisal Checklist** appropriate to the study design was applied. For conceptual and policy papers, a bespoke quality rubric assessed: (1) explicit statement of the research question or analytical aim; (2) transparency in methods (search strategy, selection criteria, data sources); (3) balanced representation of evidence including acknowledgment of limitations; and (4) clarity and logical coherence of argumentation. Studies were not excluded on the basis of quality alone; rather, quality appraisal scores weighted the interpretive confidence assigned to individual findings during synthesis.

2.5 Data Extraction and Analysis

Data were extracted using a standardized extraction form capturing: study metadata (authors, year, journal, country); study design; AI application domain; evaluation methods and metrics; governance and regulatory mechanisms described; implementation outcomes reported; and thematic codes assigned. Extraction was performed by the primary reviewer and verified by a secondary reviewer; discrepancies were resolved by discussion.

Analysis followed a **thematic synthesis** approach, combining inductive coding of primary findings with a deductive framework organized around the three analytical dimensions defined in the research question: (i) evaluation methods and metrics; (ii) governance and regulatory frameworks; and (iii) implementation outcomes and barriers. Results are reported narratively organized by theme, with summary tables providing an overview of the included studies and their key characteristics.

A PRISMA flow diagram is provided in Appendix ???. The review protocol was designed and reported following a modified PRISMA framework adapted for thematic synthesis of heterogeneous evidence.

3 Results

From an initial corpus of approximately 2,700 publications identified through systematic search across arXiv, PubMed, Semantic Scholar, and DBLP, 311 papers were retained after deduplication, abstract screening, and full-text

assessment against inclusion and exclusion criteria. Of these, 78 (25.1%) were designated Priority 3 (composite score ≥ 7.0) and form the primary evidence base; the remainder were retained as supporting literature. The reviewed literature reveals three overarching findings at the intersection of health AI evaluation and governance: a pervasive **ecological validity gap** in evaluation methodology; a structural **governance capacity deficit** in health systems; and a systematic **pilot-to-routine transition failure** rooted in organizational rather than technical barriers. These three findings are analytically interdependent rather than independent — the ecological validity gap generates misleading validation evidence that reduces the urgency of governance capacity-building; governance frameworks built on that evidence are structurally misaligned with real-world complexity; and that misalignment produces the pilot trap as a downstream consequence.

3.1 Evaluation Methods: The Ecological Validity Gap

3.1.1 Limited External Validation and Retrospective Bias

A consistent methodological limitation across the reviewed corpus is the predominance of single-site, retrospective evaluation designs. Studies evaluating deep learning models for medical imaging — including models for colorectal cancer microsatellite instability prediction, diabetic retinopathy screening, and chest X-ray interpretation — routinely report Area Under the Curve (AUC) values exceeding 0.85 in internal validation cohorts but show substantial performance decay when tested on external datasets from different institutions or populations [5]. This pattern — high internal validity, limited external generalizability — reflects a structural evaluation problem rather than a model-specific failure.

The CT-based MSI prediction study exemplifies this gap with quantitative precision: the authors report AUCs of 0.882 in their training cohort, declining to 0.768 in the internal validation cohort, 0.803 in external validation cohort 1, and 0.751 in external validation cohort 2 — a progressive performance degradation across increasingly distant populations that illustrates the generalizability challenge [5]. Importantly, even when external validation is attempted, it typically relies on retrospective data from the same health system, capturing little of the population diversity, data quality variation, and workflow integration complexity of real clinical environments.

3.1.2 Metric Design and Clinical Relevance

Standard AI evaluation metrics — accuracy, sensitivity, specificity, AUC — are poorly calibrated to the clinical context in which AI systems operate. **turchi2025ecological** demonstrate this with particular clarity in their study of AI-assisted oral cancer diagnosis: experienced dental specialists identifying real-world clinical diagnosis as requiring contextual information (beyond static images), longitudinal reasoning (beyond single-session tasks), and harm-asymmetric evaluation (a false-negative in cancer diagnosis has a different clinical weight than a false-positive). Standard AI benchmarks collapse these distinctions into a single accuracy number.

The TRAC-Pain protocol [8] offers a methodological counterpoint. By combining continuous wearable sensor data (physiological, sleep, and physical activity metrics) with ecological momentary assessments (EMAs) administered multiple times daily over a 12-week period, the study designs evaluation metrics that reflect the temporal dynamics of the pain experience — including pain interference, mood, and fatigue — rather than relying on point-of-care assessments alone. This represents an important step toward ecologically grounded evaluation, though prospective longitudinal designs of this kind remain rare in the literature.

3.1.3 Algorithmic Drift and Post-Market Monitoring

A critical and understudied dimension of AI evaluation is algorithmic drift: the phenomenon by which AI systems that were valid at the time of deployment gradually or abruptly degrade in performance as input data distributions shift — due to changes in patient populations, healthcare practices, laboratory equipment, or the AI system's own continuous learning updates. The FDA perspective paper [1] identifies continuous learning AI systems as requiring special regulatory attention precisely because the traditional pre-market validation paradigm cannot guarantee post-deployment performance. Yet mandatory post-market performance monitoring with standardized drift detection protocols is conspicuously absent from most regulatory frameworks reviewed.

Abulibdeh et al. [4] argue that this absence constitutes a **systemic safety gap**: patients and clinicians are provided no mechanism to know whether an AI system that was validated two years ago remains fit for clinical purpose today. The authors call for mandatory real-time monitoring requirements as a condition of market authorization, with clear escalation protocols and performance thresholds that trigger mandatory re-validation or market withdrawal.

3.2 Governance Structures: The Capacity Deficit

3.2.1 Regulatory Frameworks and their Assumptions

The FDA's authorized AI device landscape [1] and the EU AI Act both operate from a shared premise: that governance is a precondition that health systems and AI developers must satisfy before market entry. Under this model, authorization requires demonstrating safety and efficacy through premarket evaluation; post-market obligations are limited largely to adverse event reporting. Tian et al. [3] challenge this premise empirically: their 18-month implementation study reveals that the governance capacity required for sustainable AI deployment is not, in fact, pre-existing in health systems. Accountability boundaries, coordination mechanisms, and infrastructure governance structures emerged through implementation — they were discovered, not pre-installed.

This finding has a direct implication for regulatory design: a governance framework that treats capacity as a precondition rather than an outcome of the implementation process will systematically misidentify the point of intervention. Regulators specifying what governance should achieve without attending to whether health systems have the organizational infrastructure to deliver it will produce well-intentioned but structurally ineffective governance standards.

3.2.2 The Pilot Trap as an Organizational Failure

The pilot trap — defined as the failure of clinical AI applications to transition from short-term demonstration projects to sustained components of routine clinical care — appears across multiple health system contexts and AI application types. Tian et al. [3] attribute this not to algorithmic underperformance but to three organizational deficits: (i) unclear accountability boundaries (who is legally and professionally responsible for an AI-assisted clinical decision?); (ii) absent coordination mechanisms (how do clinical, technical, and administrative units interact around an AI system?); and (iii) inadequate infrastructure governance (how are data pipelines, model updates, and system integrations managed operationally?).

These organizational deficits are not unique to a single health system or national context. Kéri [2] documents analogous challenges in psychiatric AI deployment, where the risks of simulated empathy, illusion of attachment, and unpredictable crisis-mode behavior by AI chatbots require governance mechanisms — including strict validation protocols, continuous monitoring, and clear human accountability — that most psychiatric services are not currently equipped to provide.

3.2.3 Emerging Adaptive and Participatory Governance Models

In response to the limitations of static pre-market regulatory models, several papers in the reviewed corpus describe adaptive governance approaches that embed learning and iteration into the regulatory process itself. Regulatory sandboxes — already adopted in fintech and now proposed for health AI — allow AI systems to be deployed in controlled real-world settings with enhanced monitoring, time-limited authorization, and mandatory performance thresholds that must be met for full market entry [5, 9]. The WHO's ethics and governance guidance on AI in health similarly advocates for continuous compliance mechanisms that evolve with the AI system's performance history rather than treating authorization as a one-time gatekeeping event.

Participatory governance mechanisms — incorporating affected communities, patient advocacy groups, and frontline clinical staff into governance processes — are proposed across multiple reviewed papers as essential for AI legitimacy and sustainability [5, 6]. This aligns with the broader literature on democratic AI and public interest technology, which holds that the epistemic diversity brought by participatory processes improves both the quality and the legitimacy of governance decisions.

3.3 Implementation Outcomes: Barriers and Societal Impact

3.3.1 Organizational, Technical, and Social Barriers

The implementation barriers documented in the reviewed literature can be organized into three analytically distinct but interacting categories.

Organizational barriers include: workforce trust deficits (clinicians skeptical of AI recommendations that conflict with their clinical judgment); misaligned incentive structures (AI systems optimized for metrics that do not reflect clinical priorities or institutional goals); change management failures (inadequate training, poor human-AI workflow integration); and institutional governance deficits (absence of clear ownership, accountability, or escalation protocols for AI-assisted decisions) [3, 5].

Technical barriers include: data quality and interoperability problems (AI systems trained on curated research data that does not reflect the noise, incompleteness, and heterogeneity of real Electronic Health Record [EHR] data);

infrastructure limitations (insufficient computing resources, network reliability, or system integration capacity in resource-constrained settings); and model update challenges (managing the regulatory, technical, and organizational complexity of updating a deployed AI system) [1, 4].

Social barriers include: algorithmic literacy gaps among clinicians and patients; power dynamics that privilege technical AI teams over frontline clinical expertise; and the risk of deskilling — where reliance on AI systems erodes the development of clinical skills that are needed when AI is unavailable or fails.

3.3.2 Algorithmic Bias, Equity, and Differential Performance

The equity dimension of health AI implementation receives significant attention across the reviewed corpus. Abulibdeh et al. [4] document that AI-enabled medical devices authorized by the FDA have historically underrepresented minority populations in their training and validation data, creating a systematic risk that AI systems perform differentially across racial, socioeconomic, and geographic subgroups. This bias amplification risk is particularly acute in clinical domains such as dermatology, radiology, and cardiovascular risk prediction, where training data skews toward Caucasian and male populations. The absence of mandatory subgroup performance reporting in current regulatory frameworks means that differential AI performance may go undetected for years.

Pham [5] argue that addressing algorithmic bias requires not only technical interventions (diverse data collection, bias auditing) but also governance mechanisms that institutionalize equity considerations throughout the AI lifecycle — from problem framing and dataset construction to deployment monitoring and post-market evaluation.

3.3.3 Evidence Quality and Research Gaps

A consistent finding across the reviewed literature is the limited high-quality evidence on real-world clinical impact of AI systems. The majority of published studies report technical performance metrics (accuracy, AUC) in controlled settings. Prospective, multicenter studies measuring patient outcomes (mortality, morbidity, length of stay, quality of life) attributable to AI-assisted care remain rare. This evidence gap is compounded by publication bias: AI systems that perform well in validation studies are more likely to be published, while failed deployments are rarely reported, creating a systematically distorted literature that overstates real-world effectiveness.

The three thematic findings are analytically connected rather than parallel. The ecological validity gap does not merely reflect a technical measurement problem — it has direct consequences for governance. When AI systems are validated under conditions that do not reflect real clinical complexity, governance frameworks based on those validations will be structurally misaligned with the phenomena they seek to govern. Similarly, the governance capacity deficit is not simply a matter of underinvestment — it is partly a consequence of evaluation frameworks that generate overconfident validation evidence, reducing the perceived urgency of building pre-deployment governance infrastructure. The pilot trap can then be understood as the downstream consequence of these upstream failures: systems validated on inadequate evidence, governed by frameworks that assume absent capacity, predictably fail to transition to routine care. This interdependency implies that improving health AI outcomes requires simultaneous action across all three dimensions — better evaluation methodology, governance capacity investment, and implementation support — rather than sequential approaches that treat evaluation, governance, and implementation as separate phases.

4 Discussion

This discussion interprets the synthesized findings presented in Section 3 through an integrated analytical lens, examines their implications for research, policy, and clinical practice, acknowledges the limitations of this review, and concludes with a forward-looking research agenda identifying the most pressing unanswered questions at the intersection of health AI evaluation and governance.

4.1 Summary of Key Findings

The three thematic findings presented in the results are analytically interdependent rather than independent. The **ecological validity gap** in evaluation methodology does not merely reflect a technical measurement problem — it has direct consequences for governance. When AI systems are validated under conditions that do not reflect real-world clinical complexity, governance frameworks based on those validations will be structurally misaligned with the phenomena they seek to govern. Similarly, the **governance capacity deficit** documented in health systems is not simply a matter of underinvestment — it is partly a consequence of evaluation frameworks that generate overconfident validation evidence (suggesting AI systems are more ready for deployment than they are), which reduces the perceived urgency of building pre-deployment governance infrastructure. Finally, the **pilot trap** can be understood as the downstream consequence of these two upstream failures: systems validated on inadequate evidence, governed by frameworks that assume absent capacity, predictably fail to transition to routine care.

This interdependency implies that improving health AI outcomes requires simultaneous action across all three dimensions — better evaluation methodology, governance capacity investment, and implementation support — rather than sequential approaches that treat evaluation, governance, and implementation as separate phases.

4.2 Implications

4.2.1 *Implications for AI Evaluation Research*

The reviewed literature makes clear that the field needs a paradigm shift from single-site, retrospective evaluation to prospective, multicenter, ecologically grounded validation. This requires: (i) embedding AI evaluation within real clinical workflows rather than controlled experimental setups; (ii) adopting longitudinal evaluation designs that capture algorithmic drift and performance evolution over time; (iii) developing clinical harm-asymmetric performance metrics that reflect the differential consequences of false-positive and false-negative errors in specific clinical contexts; and (iv) mandating subgroup performance reporting to detect and quantify algorithmic bias across population strata. The TRAC-Pain protocol [8] offers a methodological template that merits wider adoption and adaptation.

4.2.2 *Implications for Regulatory and Policy Practice*

The regulatory implications cut in two directions. First, current regulatory frameworks must evolve from one-time pre-market authorization toward **lifecycle-spanning governance** that mandates continuous post-market performance monitoring, requires disclosure of training data sources, and establishes enforceable equity and bias mitigation standards [1, 4]. Second, regulators must recognize that governance capacity in health systems is not pre-given and cannot be assumed — it must be actively built through implementation support, not just prescribed through standards [3]. This suggests that regulatory frameworks should incorporate implementation support mechanisms, including regulatory sandboxes, technical assistance programs, and mandatory pre-deployment organizational readiness assessments.

4.2.3 *Implications for Health Systems and Clinical Practice*

For health systems, the reviewed evidence points to the need to invest in organizational governance infrastructure as a precondition for AI deployment, not as an afterthought. This includes: establishing clear accountability structures for AI-assisted clinical decisions; building cross-functional governance committees that bring together clinical, technical, legal, and patient representatives; developing operational protocols for model updates, performance monitoring, and incident response; and investing in workforce development to build algorithmic literacy and clinical-AI workflow integration skills at scale.

4.3 Limitations

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Data and Code Availability

Transparency and reproducibility are core principles of this work. Data and code availability are described below, subject to ethical, legal, and institutional constraints.

Data availability

The data used in this study are [publicly available / available upon request / not publicly available]. If publicly available, specify the repository and license: *“The de-identified dataset used in this study is available at [repository link] under [license].”*

If data cannot be shared publicly, state the reason and access conditions: *“Patient-level data are protected under institutional and ethical approvals and cannot be shared publicly. Qualified researchers may request access through [data access committee or institution], subject to appropriate approvals.”*

Where applicable, indicate whether aggregated or synthetic data are available.

Code availability

Code used for data preprocessing, analysis, and model training is [publicly available / partially available / not publicly available].

If available, provide a repository link and version reference: *“Code is available at <https://github.com/yourrepo/yourproject> (commit abc123) for reproducibility.”*

If parts of the code cannot be shared, briefly state which components are unavailable and why (e.g., proprietary deployment interfaces), and indicate whether pseudocode or detailed methodological descriptions are provided instead.

A Supplementary methods

Include detailed methodological information that would disrupt the flow of the main text. Present additional analyses or derivations.

B Additional results

Present supporting analyses and figures that complement your main results. Include any results referenced but not shown in the main text.